HOMEWORK 5.1 · UNIT 5 · NC.8.G.7 · NC.8.G.6 · NC.8.EE.2

Pythagorean Theorem (Intro)

Jan 5 · Targets: G.7-3, G.7-1, G.6-2, EE.2-2, G.6-1, G.7-5, G.6-7, G.6-5, G.6-8, G.7-7

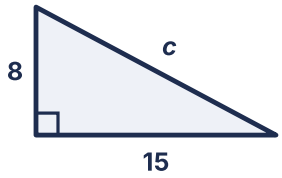
Name _____ Date _____ Block _____

Remember: In a right triangle, $a^2 + b^2 = c^2$. a and b are the **legs** (they make the right angle). c is the **hypotenuse** (the longest side, across from the right angle). Example: $3^2 + 4^2 = 9 + 16 = 25$, so $c = \sqrt{25} = 5$.

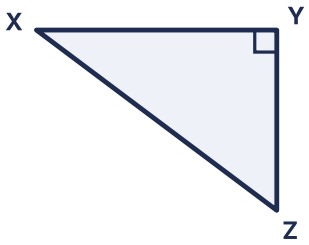
SCORE 3.0 Approaching · the basics

Targets: G.7-3, G.7-1, G.6-2, EE.2-2, G.6-1

1 Find c. G.7-3

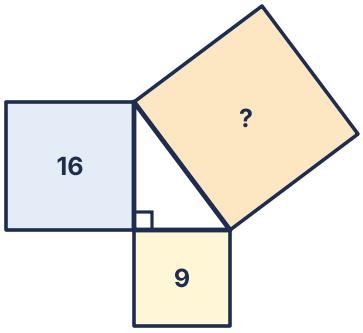


2 Which side is the **hypotenuse**? G.7-1



Ⓐ XY Ⓑ YZ Ⓒ XZ

3 A square is built on each side. Find the area of the third square. Then find c. G.6-2



4 Find $\sqrt{121}$. REVIEW 1.3 EE.2-2

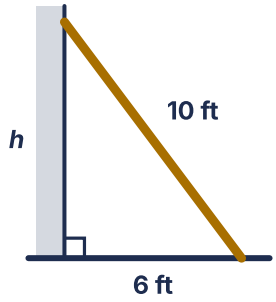
5 The legs are the two sides that make the right angle. G.6-1

True False

SCORE 4.0 Proficient · on target

Targets: G.7-5, G.6-7, G.6-5

6 A 10-ft ladder leans on a wall. Its bottom is 6 ft from the wall. How high up the wall does it reach? G.7-5



7 **Marcus** made a mistake. Circle it. Explain it. Fix it. G.6-7

MARCUS'S WORK

Legs 5 and 12: $5^2 + 12^2 = c^2$

$25 + 144 = 169$

$c = 169$

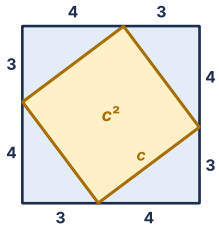
8 The squares on the legs of a right triangle have areas 81 and 144. What is the area of the square on the hypotenuse? G.6-5

Ⓐ 225 Ⓑ 63 Ⓒ 15 Ⓓ 21

SCORE 5.0 Mastery · go deeper

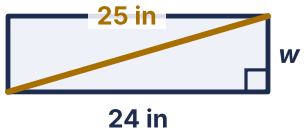
Targets: G.6-8, G.7-7

9 The big square is 7 by 7. Each triangle has legs 3 and 4. Use areas to show that the tilted square has area $3^2 + 4^2$. G.6-8



Big square = _____ . Triangles = _____ . Tilted square = _____ .

10 A rectangle is 24 in long. Its diagonal is 25 in. Find the perimeter. Explain. G.7-7



First I find _____ . Then _____ .